

Dengue Vector SDM Workshop: Setup Guide

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Welcome! In this workshop we will use R to build a first species distribution model for the dengue vector *Aedes aegypti* in Brazil.

The workshop is beginner-friendly, but it does rely on several R packages. Please try to complete this setup before the session. If something fails, do not worry: we will have cached data available during the workshop.

1. Install R

Download and install the latest version of R from:

<<https://cran.r-project.org/>>

Choose the version for your operating system:

- Windows: "Download R for Windows"
- macOS: "Download R for macOS"

2. Install RStudio Desktop

Download and install RStudio Desktop from:

<<https://posit.co/download/rstudio-desktop/>>

R is the programming language. RStudio is the interface we will use to write and run R code.

3. Download The Workshop Folder

You should have a folder called:

SDM_dengue_Brazil_workshop

Put it somewhere easy to find, such as your Desktop or Documents folder.

4. Open The RStudio Project

Inside the workshop folder, double-click:

```
SDM_dengue_Brazil_workshop.Rproj
```

This opens RStudio in the correct folder. This means you should not need to use `setwd()`.

5. Install The R Packages

In RStudio, open:

```
scripts/00_install_packages.R
```

Run the whole script.

The most important packages are:

```
install.packages(c(
  "biomod2",
  "gbm",
  "dplyr",
  "ggplot2",
  "readr",
  "rgbif",
  "sf",
  "terra",
  "viridis",
  "rnatrualearth",
  "rnatrualearthdata",
  "randomForest"
))
```

Optional packages:

```
install.packages(c("CoordinateCleaner", "geodata"))
```

The workshop scripts do not require these optional packages, but they are useful for future SDM work.

6. Check Your Setup

Open and run:

```
scripts/01_setup_check.R
```

If everything is working, you should see a message saying that the setup check passed.

7. Common Installation Problems

Windows

If package installation asks whether to compile from source, choose "No" unless you already have RTools installed.

macOS

Some spatial packages may ask for command line tools. If prompted, install them and try again.

Slow Internet

The workshop includes cached occurrence data and environmental layers. If the live GBIF download fails, you can still complete the model.

8. During The Workshop

The participant-facing lesson is:

`lesson.html`

The short intro slide deck is:

`slides/intro_sdm_dengue_brazil.pptx`

During the live workshop, we will mostly use one script and run it section by section:

`scripts/workshop_aedes_aegypti_brazil_sdm.R`

The other scripts are setup checks, cached-data builders, or optional exports.

Outputs will be saved into:

`outputs/`

The model is a teaching model. It estimates environmental suitability for *Aedes aegypti*, not dengue transmission or human disease risk.